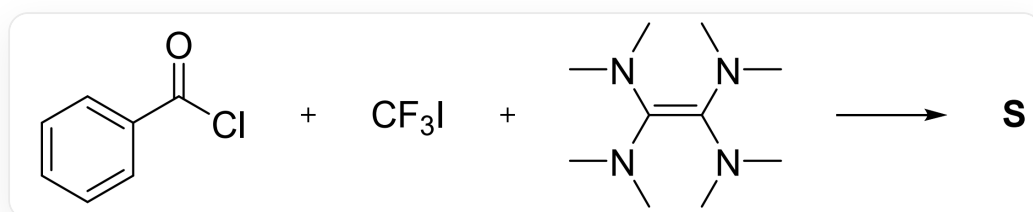


Question

Hexafluoroacetone can undergo significant hydration. In a concentrated solution of NaOH, it reacts with p-methoxybenzaldehyde at 100 °C to obtain alcohol **O** and salt **P**.

Similarly, benzoyl chloride reacts with trifluoromethyl iodide and TDAE in a theoretical 1:1:1 ratio to obtain ester **S**. It is believed that a benzene ring-containing intermediate **T** is first generated, and **T** then reacts with benzoyl chloride to obtain **S**. At the same time, salt **U** is also produced. It is known that TDAE is tetra(dimethylamino)ethylene.



C1C=CC=C(C(Cl)=O)C=1.C(F)(I)(F)F.C(N(C)C)(N(C)C)=C(N(C)C)N(C)C>>[S]

Which of the following statements are correct:

1. The chemical formula of **O** is $C_3F_6OH_2$
2. **P** does not contain hydrogen element
3. In the reaction, TDAE is oxidized to the cation of **U**
4. The mass fraction of oxygen in **S** is 9.19%
5. **S** has 3 kinds of chemically different hydrogens

A. All other options are incorrect

B. 2.3.4.

C. 1.2.4.

D. 1.2.3.4.5

E. 2.3.4.5.

F. 1.3.5.

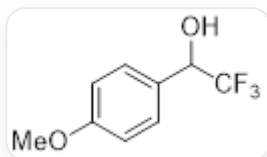
G. 2.3.

Answer

Correct Answer: **B**

Detailed Explanation

The problem states that hexafluoroacetone can undergo significant hydration, and in a concentrated solution of NaOH, it can undergo trifluoromethyl anion transfer reaction to obtain the compound **O** and sodium trifluoroacetate (salt **P**) as shown in the figure.



OC(C(F)(F)F)C1=CC=C(OC)C=C1

CHECKPOINT

1 PTS

O is OC(C(F)(F)F)C1=CC=C(OC)C=C1

CHECKPOINT

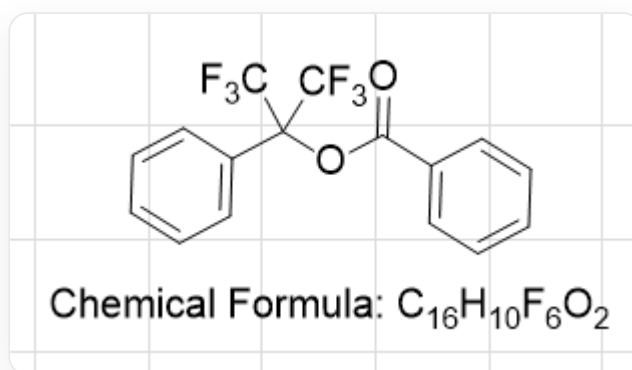
1 PTS

P is CF3COONa

Therefore, 1 is incorrect, and 2 is correct.

In the second reaction, TDAE can reduce CF_3I to obtain CF_3^- equivalent to attack acyl chloride twice to obtain the tertiary alcohol intermediate **T**: OC(C(F)(F)F)(C(F)(F)F)C1=CC=CC=C1. At the same time, the cation TADF^{2+} of **U** is obtained. Afterwards, the tertiary alcohol undergoes acylation again to obtain the product **S**: O=C(C1=CC=CC=C1)OC(C(F)(F)F)(C(F)(F)F)C2=CC=CC=C2.

The structure of **S** is shown in the figure:



CHECKPOINT

1 PTS

T is OC(C(F)(F)F)(C(F)(F)F)C1=CC=CC=C1

CHECKPOINT

1 PTS

S is O=C(C1=CC=CC=C1)OC(C(F)(F)F)(C(F)(F)F)C2=CC=CC=C2

Therefore, 3 and 4 are correct (based on the correct mass fraction of oxygen in the chemical formula), 5 should be 6 types, which is incorrect. Option B is correct.